```
Using device: cpu
Using single-amplitude feyn data from /Users/kymani/Documents/New_Scattering_AI/Repo/last_predictions/ym_4pt/03_constructed_inputs/gluon4feyn1234_model_ready.csv
Imported 1 token rows
Wrote raw expression data to /private/tmp/instruction-eval-check-yylw3nki/ym_4pt/eval.csv
Wrote normalized token data to /private/tmp/instruction-eval-check-yylw3nki/ym_4pt/eval_tok.csv
Input token limit: unlimited; longest input: 1247 content / 1249 with BOS/EOS
Numerical evaluator: ym (energy_scale=2.0; pol_modes=coulomb,covariant; tol_abs=1e-10; tol_rel=1e-08)
Loading model from /Users/kymani/Documents/New_Scattering_AI/Repo/last_predictions/ym_4pt/model/best_model.pt

Running decode mode: greedy; max_length=160; beam_size=1; mask_invalid_tokens=False
  wrote human CSV to /private/tmp/instruction-eval-check-yylw3nki/ym_4pt/eval_greedy_human.csv

[greedy]  n=1
  top-1 equivalent   :    1 / 1  (100.0%)
  any-beam equivalent:    1 / 1  (100.0%)
  avg valid beams : 1.0
  shorter equivalent outputs: 1 / 1
  copied inputs: 0 / 1
  independent target: unavailable; exact-target metrics are not applicable
  avg tokens (correct):  pred 87.0  vs  scrambled 1247.0

Example predictions (1 rows):
  row 0 [greedy]
    target simple :
    input scram   : (((e_1 · e_2)*(e_3 · e_4)*(p_1 · p_3))/(p_1 · p_2)) + (((e_1 · e_2)*(e_3 · p_1)*(e_4 · p_1)*(p_3 · p_4))/((p_1 · p_2)*(p_1 · p_4))) − ((((e_1 · e_2)*(e_3 · p_1)*(e_4 · p_3))/(p_1 ·
    top1 pred     : (p_1·F_2·F_3·p_1*p_2·F_1·F_4·p_2)/(((p_1·p_2))^2*p_1·p_3*p_1·p_4) + (p_1·F_2·F_4·p_1*p_2·F_1·F_3·p_2)/(((p_1·p_2))^2*((p_1·p_4))^2) + (p_1·F_3·F_4·p_1*p_3·F_1·F_2·p_3)/(p_1·p_2*p_1
    top1 metrics  : token=None string=None num_simple=None num_scr=1
    any-beam      : token=None string=None num_simple=None num_scr=1
  wrote detail results to /private/tmp/instruction-eval-check-yylw3nki/ym_4pt/eval_greedy_results.csv

Wrote summary to /private/tmp/instruction-eval-check-yylw3nki/ym_4pt/eval_summary.csv
Total wall time: 2.95s

Terminal output PDF: /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testing/outputs/Yang-mills_4_Evaluation_Information_2026-09-29_15-14-06-441401.pdf
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